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Roll No

CS-8304 (GS)
B.E. VIII Semester Examination, June 2020
Grading System (GS)
Quantum Computing
Time : Three Hours

Maximum Marks : 70

Note: i) Attempt any five questions.
ii) All questions carry equal marks.

1. Discuss about the computational complexity of Discrete Fourier Transform.
2. Explain graver's search alogorithm.
3. A system is found to be in the state

$$|\psi\rangle = \frac{1}{\sqrt{5}}|0\rangle + \frac{2}{\sqrt{5}}|1\rangle$$

- a) Write down the density operator for this state.
 - b) Write down the matrix representation of the density operator in the $\{|0\rangle, |1\rangle\}$ basis. Verify that $\text{Tr}(\rho) = 1$ and show this is a pure state.
4. a) Discuss Bell's theorem in Quantum computing.
b) Explain the application of entanglement in teleoperation.
 5. a) Explain controlled and composite gates.
b) Explain density operator for a pure state.
 6. Explain the following.
a) Shannon Entropy
b) Data compression
 7. Suppose that $f(00) = f(01) = 0$, $f(10) = f(11) = 1$. Apply the Deutsch-Josza algorithm and show that atleast one of the first two Qubits ends up as a 1.

OR

Explain Harmonic Oscillator Quantum Computer and Optical Photon Quantum Computer.

8. Answer any four of the following:
a) Are Quantum gates Reversible or Irreversible? Explain.
b) Discuss the properties of Density Operator.
c) Discuss the conditions of Quantum Computation.
d) How serial and parallel operations are represented.
e) What is Stabilizer Code? Explain.
f) What are the requirements of Fault Tolerance?
